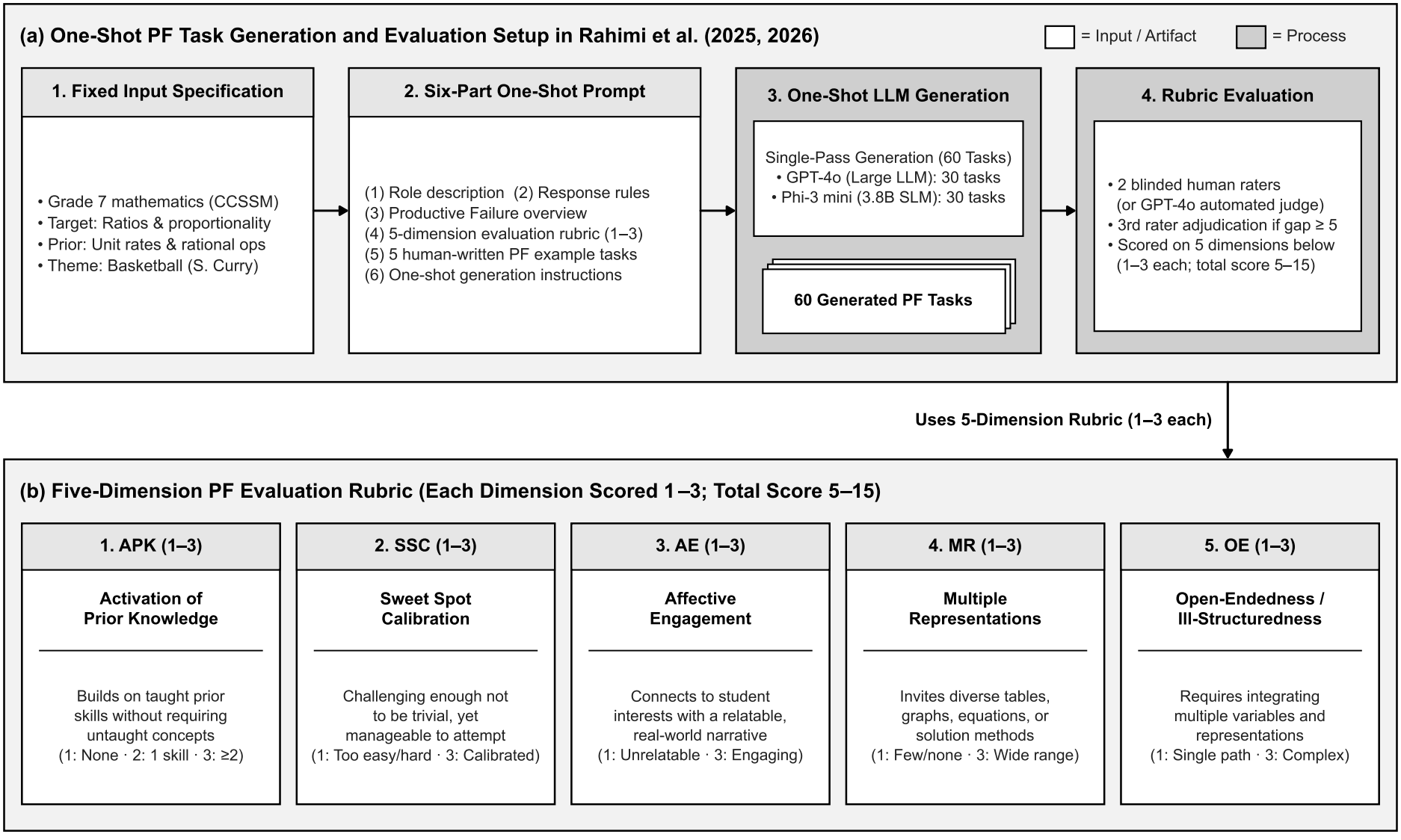
(a) One-Shot PF Task Generation and Evaluation Setup in Rahimi et al. (2025, 2026)
= Input / Artifact
= Process
1. Fixed Input Specification
• Grade 7 mathematics (CCSSM)
• Target: Ratios & proportionality
• Prior: Unit rates & rational ops
• Theme: Basketball (S. Curry)
2. Six-Part One-Shot Prompt
(1) Role description (2) Response rules
(3) Productive Failure overview
(4) 5-dimension evaluation rubric (1–3)
(5) 5 human-written PF example tasks
(6) One-shot generation instructions
3. One-Shot LLM Generation
Single-Pass Generation (60 Tasks)
• GPT-4o (Large LLM): 30 tasks
• Phi-3 mini (3.8B SLM): 30 tasks
60 Generated PF Tasks
4. Rubric Evaluation
• 2 blinded human raters
(or GPT-4o automated judge)
• 3rd rater adjudication if gap ≥ 5
• Scored on 5 dimensions below
(1–3 each; total score 5–15)
Uses 5-Dimension Rubric (1–3 each)
(b) Five-Dimension PF Evaluation Rubric (Each Dimension Scored 1–3; Total Score 5–15)
1. APK (1–3)
Activation of Prior Knowledge
Builds on taught prior skills without requiring untaught concepts
(1: None · 2: 1 skill · 3: ≥2)
2. SSC (1–3)
Sweet Spot Calibration
Challenging enough not to be trivial, yet manageable to attempt
(1: Too easy/hard · 3: Calibrated)
3. AE (1–3)
Affective Engagement
Connects to student interests with a relatable, real-world narrative
(1: Unrelatable · 3: Engaging)
4. MR (1–3)
Multiple Representations
Invites diverse tables, graphs, equations, or solution methods
(1: Few/none · 3: Wide range)
5. OE (1–3)
Open-Endedness / Ill-Structuredness
Requires integrating multiple variables and representations
(1: Single path · 3: Complex)